Dataset Description and Preprocessing Report

CS Vision – Data Analytics Internship Program (Set A) Project: Financial & Operational Performance Analytics System Week 1: Data Collection and Preprocessing (20–26 October 2026) Prepared by: Tanisha Sinha

1. Objective

The goal of Week 1 is to collect a suitable financial and operational dataset, assess its quality, clean it, and produce an analytical dataset that supports the revenue, expense, profit, regional, product and monthly trend analysis planned for Weeks 2–8.

2. Dataset Description

Source: Global Superstore dataset, Kaggle (`apoorvaappz/global-super-store-dataset`), file `Global_Superstore2.csv` . It contains real-world-style retail order records from a global retailer.

Size: 51,290 order lines × 24 columns, far above the 500+ record recommendation.

Coverage: 1 January 2011 to 31 December 2014 (4 years), 7 markets, 13 regions, 147 countries, 3 product categories, 17 sub-categories, 1,590 customers, 3 customer segments and 25,035 distinct orders.

Brief category	Columns in the dataset
Transaction	Row ID, Order ID, Order Date, Ship Date
Revenue	Sales
Expense	Shipping Cost (derived: Cost)
Profit	Profit (derived: Net Profit)
Product	Product ID, Product Name, Category, Sub-Category
Customer	Customer ID, Customer Name, Segment
Location	City, State, Country, Postal Code, Market, Region
Operations	Quantity, Discount, Ship Mode, Order Priority
Time	Derived from Order Date: Year, Quarter, Month

3. Initial Data Profiling

Check	Result
Duplicate rows	0 (Row ID is unique)
Missing values	Only Postal Code: 41,296 of 51,290 rows (80.5%)
Date format	DD-MM-YYYY text; all Order Dates and Ship Dates parsed, 0 failures
Ship Date earlier than Order Date	0 rows
Sales ⩽ 0 or Quantity ⩽ 0	0 rows
Loss-making lines (Profit < 0)	12,544 (24.5%)
File encoding	Not UTF-8; loaded with Latin-1

Postal Code is present for 9,994 rows. These are the United States orders, since the field does not apply to most other countries. The 41,296 missing values are structural, not data entry errors.

No explicit Order Status or Payment Status column exists in this dataset.

4. Preprocessing Methodology

Step 1 – Loading. The file was read with Latin-1 encoding. Column names were standardised (spaces and hyphens replaced with underscores, for example `Order Date` to `Order_Date`).

Step 2 – Duplicates. A duplicate check found none, so no rows were removed.

Step 3 – Text standardisation. Leading and trailing spaces were removed from all text columns. The categorical fields (Segment, Market, Region, Category, Ship Mode, Order Priority) were inspected and found to use consistent labels.

Step 4 – Type conversion. Order Date and Ship Date were converted from text to datetime. Sales, Profit, Quantity, Discount and Shipping Cost were confirmed numeric. Postal Code was converted to text because it is an identifier, not a quantity.

Step 5 – Missing values. The 41,296 missing Postal Codes were labelled "Unknown". They were not imputed, because postal codes cannot be estimated and the field is not used in the financial analysis.

Step 6 – Validity checks. No rows had zero or negative Sales or Quantity, so none were removed. The 12,544 negative-profit lines were **kept**, because they are genuine business losses that matter for the profitability analysis in Weeks 3 and 4.

Step 7 – Derived financial fields.

Field	Formula	Reason
Cost	Sales − Profit	The dataset has no cost column
Gross Profit	Profit	Profit as recorded
Operating Expense	Shipping Cost	An operational expense
Net Profit	Gross Profit − Operating Expense	Profit after shipping expense
Gross and Net Margin %	Profit ÷ Sales × 100	Margin KPIs
Cost to Revenue	Cost ÷ Sales	Cost-ratio KPI
Year, Quarter, Month, Month Name, Year-Month, Day of Week, Shipping Days	From Order Date and Ship Date	Time analysis

Step 8 – Outliers. An IQR rule (3 × IQR, applied within each sub-category) flagged 1,605 Sales outliers and 3,106 Profit outliers. In retail these are mostly legitimate large orders or heavy discounting, so they were flagged for the Week 3 anomaly analysis and were **not** removed or altered.

5. Validation Results

All checks passed on the final dataset: no missing values, Order Dates parsed, Sales and Quantity positive, Cost = Sales − Profit, Net Profit = Gross Profit − Operating Expense, and valid Quarter and Month values.

Metric	Raw	Clean
Rows	51,290	51,290
Columns	24	40
Missing cells	41,296	0
Duplicate rows	0	0

The 16 new columns are the derived financial, time and outlier-flag fields described above.

6. NumPy Analysis (Week 1 Focus)

NumPy was used for array creation, arithmetic, descriptive statistics, conditional logic, grouped aggregation and a correlation matrix.

- Total sales: 12,642,502. Mean order-line sales 246.49, median 85.05, standard deviation 487.57. The mean far exceeds the median, so the distribution is strongly right-skewed.
- Total gross profit: 1,467,457 (11.6% of sales). Shipping cost totals 1,352,821, so total net profit is only 114,637 (0.91% of sales).
- 21,150 order lines (41.2%) lose money after shipping, with a combined loss of 1,366,439.
- Discounts hurt profit. Lines with a discount above 20% average a net **loss** of 89.92 per line, while undiscounted lines average a net profit of 35.34. Discount and Profit are negatively correlated (r = −0.32).
- Sales by year grew each year: 2.26M (2011), 2.68M (2012), 3.41M (2013), 4.30M (2014).
- Sales by category: Technology 4.74M, Furniture 4.11M, Office Supplies 3.79M.
- Net margin varies sharply by region. The best are Canada (15.6%), North Asia (8.3%) and Central Asia (7.3%). The worst are Southeast Asia (−8.6%), EMEA (−5.5%) and South (−1.8%).

These are preliminary observations. Full analysis follows in Week 2 and Week 3.

7. Assumptions and Limitations

- **Cost and Net Profit are derived.** Cost = Sales − Profit assumes the recorded Profit is before shipping. If the source already deducts shipping, Net Profit is understated. This assumption should be stated when presenting results.
- There is no Order Status or Payment Status column. Ship Mode and Order Priority are used to represent operational status.
- Monetary values have no currency stated in the source and are reported as given.
- Outliers are flagged but retained, so Week 3 anomaly findings may include legitimate large orders.

8. Deliverables Produced

File	Description
clean_financial_data.csv / .xlsx	Clean analytical dataset (51,290 rows × 40 columns)
Week1_Data_Preprocessing_Superstore.ipynb	Documented preprocessing notebook
This report	Dataset description and preprocessing report

9. Conclusion

The dataset was already structurally clean. Its main issues were a text date format, a non-UTF-8 encoding and a structurally empty Postal Code field. Each was handled with a documented rule. Cost, Net Profit and margin fields were derived, and the analysis-ready dataset passed all validation checks. Preliminary NumPy results show healthy sales growth but thin net margins, driven by shipping costs and discounting, which sets up the profitability work in Weeks 2–4.